

CENTRAL EUROPEAN CATEGORY THEORY SEMINAR

BRNO, CZECH REPUBLIC—20 NOVEMBER, 2026

PROGRAMME

10	**Meeting begins**
10.10-11	Nils Carqueville
11-11.50	Tomáš Jakl
12-13.30	**Lunch**
13.45-14.40	Niels van der Weide
14.40-15	**Coffee break**
15-15.10	Jiří Adámek
15.10-15.20	Ödül Tetik
15.20-15.30	Dominik Trnka
15.30-15.40	Chun-Yu (Max) Lin
15.40-15.50	Miloslav Štěpán
15.50-16	Aleksandar Ivanov
16-16.30	**Coffee break**
16.30-16.40	Pavla Procházková
16.40-16.50	Quentin Aristote
16.50-17	Szumi Xie
17-17.10	Maximilian Strohmeier
17.10-17.20	Giuseppe Leoncini
17.20-17.30	Kristóf Kanalas
17.30	**Closing remarks**
19-	**Dinner at Castellana Trattoria**

TITLES AND ABSTRACTS

Invited Talks

Carqueville, Nils: Higher categories in topological quantum field theory

Abstract: Topological quantum field theories (TQFTs) are symmetric monoidal functors on bordism categories, whose objects and morphisms are roughly (n-1)-dimensional manifolds and n-dimensional manifolds with boundary for some positive integer n. I'll give details on the definition, explain the motivation from physics to study such functors, and explain how to extract higher categories from them.

Jakl, Tomáš: Categorical finite model theory: game comonads and arboreal categories

Abstract: Game comonads and arboreal categories are a novel branch of categorical logic, establishing a connection between category theory and finite model theory. The latter is a branch of mathematical logic focusing on logics designed to solve algorithmic and complexity theoretic questions and to establish expressibility of various computational systems (e.g. database queries).

However, the logics studied in finite model theory often lack essential properties. For example, they often lack compactness and completeness (due to the theorem of Trakhtenbrot). Consequently, traditional approaches of categorical logics, such as the toolkit of locally presentable and accessible do not directly apply to this setting. We describe game comonads and arboreal categories, which exploit the fact that logical equivalences (not only in finite model theory) are characterised by model-theoretic games, such as the Ehrenfeucht-Fraïssé and bisimulation games. Intuitively, game comonads describe the unravelling given by the rules of these games and arboreal categories describe synthetic properties of categories where these unravellings live (typically these are the Eilenberg–Moore coalgebras of game comonads).

In this talk I overview the theory of game comonads and arboreal categories and describe some areas where this theory meets the theory of finitely presentable and accessible categories. I will also give a brief description of the representation theory of arboreal categories, which is a part of an ongoing work with Luca Reggio.

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van der Weide, Niels: Internal models of linear type theories

Abstract: The advancement of linear logic has witnessed its core ideas being woven into various forms of type theory, such as System F and dependent type theory. To provide general semantics for these systems, categorical structures have been proposed: linear hyperdoctrines in the case of System F and linear comprehension categories in

the case of linear dependent type theory. These notions bear a fundamental similarity in that they are variations on linear-non-linear (LNL) adjunctions.

In this talk, we make these similarities precise. Specifically, we define a general notion of linear-non-linear adjunction that unifies categorical semantics of different linear theories. The general notion is a reformulation of the familiar notion internal to 2-categories with finite products. We recover the familiar notion when the ambient 2-category is that of categories, functors and natural transformations. We also show that pseudofunctors between 2-categories preserve linear non-linear adjunctions whenever they preserve finite products. As an application, we study externalization of internal categories, demonstrating how various concrete models arise this way, including realizability models and the discrete families model.

Contributed Talks

Adámek, Jiří: Finitarity Properties of Presheaf Categories

Abstract: The category $\hat{C}$ of presheaves on a small category C is locally finitely presentable (lfp) and cartesian closed. We characterize categories C for which $\hat{C}$ is lfp as a closed category. Finite multi-products in C are sufficient for this, whereas weak finite multi-products are necessary. We further characterize categories such that every finitely generated presheaf is finitely presentable in $\hat{C}$.

Aristote, Quentin: Profunctorial algebras

Abstract: We provide a bicategorical generalization of Barr's landmark 1970 paper, in which he describes how to extend Set-monads to relations and uses this to characterize topological spaces as the relational algebras of the ultrafilter monad. With two-sided discrete fibrations playing the role of relations in a bicategory, we first describe how to extend pseudomonads on a bicategory to skew monads on its bicategory of two-sided discrete fibrations, and we characterize in terms of exact squares when these extensions are themselves pseudomonads. As a wide class of examples, we show that every Set-monad induces a pseudomonad on the 2-category of categories admitting a skew extension to profunctors, and in a few relevant cases we introduce suitable quotients also extending to profunctors. Among the latter, we then focus on the ultracompletion pseudomonad, whose pseudoalgebras are ultracategories: we characterize the normalized lax algebras of its profunctorial extension as ultraconvergence spaces, a recently-introduced categorification of topological spaces.

Ivanov, Aleksandar: Bistellar data in arbitrary dimensions

Abstract: The orbifold construction is a procedure that constructs new TQFTs from old ones given a particular categorical input. I will explain the basics of the construction,

and then I will give a framework to construct a specific n-category, such that functors out of it give the most general input data called bistellar data. The benefit of the framework is that it gives an immediate description in any dimension n, as well as a description of bistellar bimodules and all further iterated bimodules, which should serve as higher cells in a Morita-style higher category.

Kanalas, Krystóf: Realizing types is a local property

Abstract: I will recall the construction of Artin gluing, and use it to prove the following: if X is a space and $\mathbf{Sh}(X)$ can realize types, then for any open subset U the topos $\mathbf{Sh}(U)$ can realize types, and conversely, if $(U_i)_i$ is an open cover of X and each $\mathbf{Sh}(U_i)$ can realize types, then $\mathbf{Sh}(X)$ can realize types. The analogous claim for arbitrary frames also holds, and it gives new examples of Grothendieck toposes which realize types.

This is part of a joint work with Rodrigo Nicolau Almeida.

Leoncini, Giuseppe: A double categorical approach to weakly enriched category theory.

Abstract: A formal category theory is constructed (in the form of a proarrow equipment), encoding weak coherent enrichment over a monoidal model category $\mathcal{V}$. We describe how basic categorical concepts formulated via the equipment translate back to enriched categories. We characterize Dwyer-Kan equivalences of enriched categories as 2-categorical equivalences.

Specializing to either the Kan-Quillen model structure on simplicial sets, or the Quillen-Serre model structure on topological spaces, we prove that the resulting formal category theory is equivalent to the one associated with the ∞-cosmos of quasicategories, thereby extending the formal approach to $(\infty, 1)$-categories in the sense of Riehl-Verity to encompass both simplicial and topological categories.

Lin, Chun-Yu (Max): On enriched regular-linear theories.

Procházková, Pavla: Partial composition of operads as a skew monoidal product

Abstract: Operads and multicategories formalise the notion of multimaps – arrows that take multiple inputs and produce a single output. There are two ways to treat composition of these multimaps: given an n-ary map g, we can either compose it with an n-tuple of multimaps $(f_1, ..., f_n)$, or compose it with a single multimap f_i at a time, at a specified position in the domain of g.

The first approach can be captured by considering the substitution monoidal product and taking monoids internal to it. In this speed talk, we explore how to capture operads

with partial composition in an analogous way. Fiore proposed, and Obradović further elaborated on, the idea that the appropriate structure in which to take monoids is a certain monoidal structure warped by a derivative functor. Taking the derivative roughly corresponds to specifying the position at which composition takes place. This new product fails to be monoidal, but instead constitutes an instance of a structured skew monoidal product.

Štěpán, Miloslav: Turning lax algebras into strict ones using Kleisli objects

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Strohmeier, Maximilian: Categorical Ramsey Theory: A Brief Glance

Abstract: Introduction to categorical Ramsey theory, providing basic definitions and highlighting a few results. In particular, classification of Ramsey statements using presheaf properties and a categorical partite construction.

Tetik, Ödül: Unleashing stratified spaces on the rest of category theory

Abstract: Stratified spaces are typically poset-stratified, so their homotopy types correspond to infinity-categories that look roughly like posets. I'd like to outline an idea to stratify spaces over categories instead, and briefly discuss some potential directions one could explore.

Trnka, Dominik: Lax distributivity of monad algebras

Xie, Szumi: Groupoid categories with families
